

Degree: Let $X \subseteq \mathbb{P}^n$ be a variety. Its degree is the number of intersection points w/ a generic linear subspace of dimension $n - \dim X$.

E.g: Bezout's theorem $\Rightarrow$ the degree of a curve

$V(f) \subseteq \mathbb{P}_{\mathbb{C}}^2$ is the degree of $f \in \mathbb{C}_1[x_0, x_1, x_2]$

generic = "almost all"

Example 3.3.3: $A_1 := \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 2 \end{pmatrix}$

$\psi_{A_1}: (\mathbb{C}^*)^2 \rightarrow \mathbb{P}^5$

$(t_1, t_2) \mapsto (1 : t_1 : t_2 : t_1^2 : t_1 t_2 : t_2^2)$

second Veronese embedding of $\mathbb{P}^2$

ψ_{A_1} has 6 generators and degree 4 } (see Synter-NoteBook)

👍: Computing degree of proj. toric varieties is "easy", because of next result.

Kushnirenko's Theorem (3.3.1): Let $A \in \mathbb{Z}^{d \times n}$ be such

that $\mathbb{Z}^n A = \mathbb{Z}^d$ (Recall $\mathbb{Z}^n A := \{ \sum_{i=1}^n \lambda_i a_i : \sum \lambda_i = 0, \lambda_i \in \mathbb{Z} \}$)

Then $\deg(X_A) = d! \text{Vol}(A)$

Remarks: $d! \text{Vol}(A)$ is the NORMALIZED VOLUME. It counts the number of "unit" simplices that can fit into $\text{Conv}(A)$.



both simplices have normalized volume 1, and P has normalized volume $2 \sqrt{}$

• Hilbert function theorem | $X \subseteq \mathbb{P}^n$ projective variety, $\dim(X) = d$.
 Let $\mathbb{C}[X] = \mathbb{C}[x_0, \dots, x_n] / I(X)$ its coordinate ring. Recall it has a grading given by

$$\mathbb{C}[X] = \bigoplus_{k \geq 0} \frac{\mathbb{C}[x_0, \dots, x_n]_k}{I(X)_k}$$

($\mathbb{C}[x_0, \dots, x_n]_k$ is the $\mathbb{C}$ vector space of homog. polynomials of degree k . Similarly $I(X)_k$).

⚠ $I \subseteq \mathbb{C}[x_0, \dots, x_n]$ homogeneous $\Leftrightarrow I = \bigoplus_{k \geq 0} I_k$, this is why the grading makes sense)

Let $HF_X: \mathbb{Z} \longrightarrow \mathbb{N}$ be its HILBERT FUNCTION
 $k \longmapsto \dim_{\mathbb{C}}(\mathbb{C}[X]_k)$

Then there exists a polynomial $HP_X(k) \in \mathbb{C}[k]$, called HILBERT POLYNOMIAL, s.t. For $k \gg 0$ sufficiently large, we have

$$HF_X(k) = HP_X(k) \quad (k \gg 0)$$

Moreover, the leading monomial of $HP_X(k)$ is

$$\frac{\deg(X)}{d!} \cdot k^d$$

□

Recall: the Ehrhart polynomial is defined by

$$E_A(k) := E_{X_A}(k) = |(k \cdot \text{Conv}(A)) \cap \mathbb{Z}^d| \quad \text{and has leading term } \text{Vol}(A) \cdot k^d.$$

Claim: The leading terms of E_A and HP_A agree.

! $E_A(k)$ and $HF_A(k)$ are different in general

Recall (prop 2.1.10)

$$\mathbb{C}[X_A] \stackrel{\text{grad.}}{\simeq} \bigoplus_{k \geq 0} \left(\bigoplus_{\substack{m \in NA \\ m_{d+1} = k}} \mathbb{C} \cdot t^m \right), \text{ i.e. the grade}$$

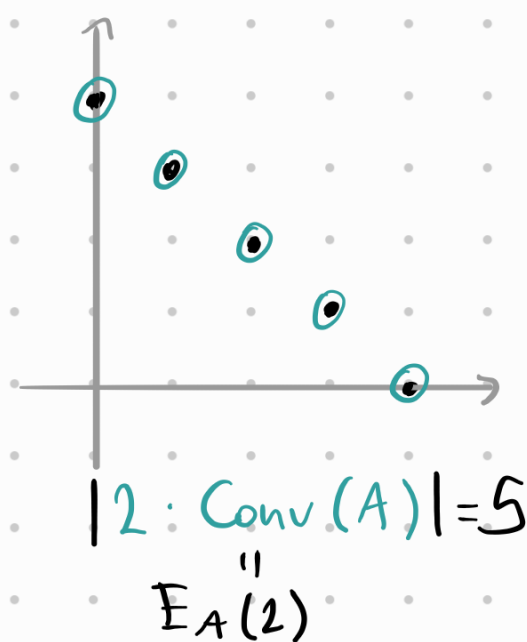
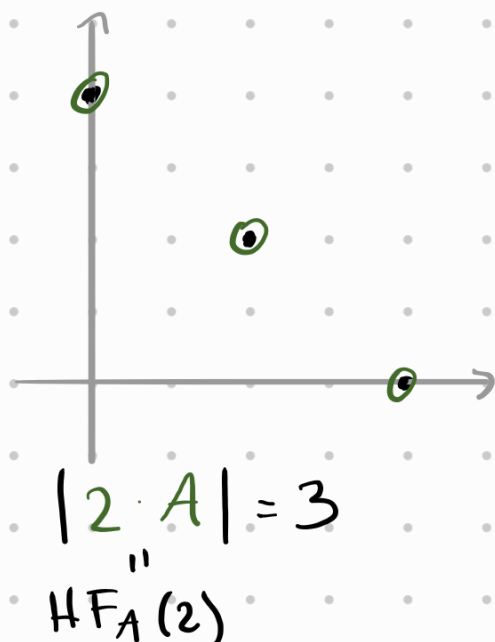
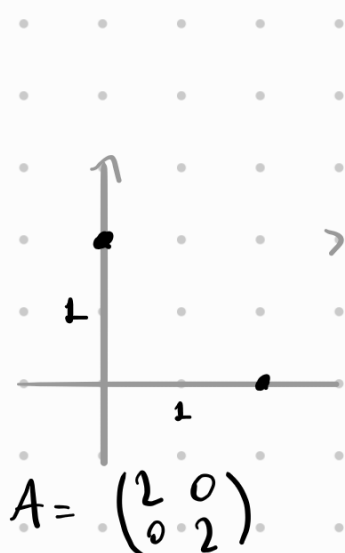
of a monomial $t^{(m_1, \dots, m_d, m_{d+1})} \in NA$ is given by the last coordinate of the exponent.

Hence,

$$\dim_{\mathbb{C}} \mathbb{C}[X_A]_k = |(k \cdot A) \cap \mathbb{Z}^n|, \text{ the number of lattice point that can be obtained by summing } k \text{ elements of } A.$$

$$\text{Hence } HF_A(k) \leq E_A(k), \forall k \geq 0.$$

In general this is not an equality:



proof of Claim: Two steps:

i) Let $\sigma^\vee = \text{Cone}(\hat{A}) \subseteq \mathbb{R}^{d+1}$, $S_\sigma := \sigma^\vee \cap \mathbb{Z}^{d+1}$

If $\mathbb{Z}^1 A = \mathbb{Z}^d$, then $\exists m \in M\hat{A}$ s.t. $m + S_\sigma \subseteq M\hat{A}$
(lemma 3.3.8)

ii) By i), $\exists \tilde{m} \in \mathbb{Z}^d$, $e \in M$ s.t. $(\tilde{m}, e) + S_\sigma \subseteq M\hat{A}$.

Observe that, because of inclusions:

$$\begin{aligned} |\underline{(k-e)\text{Conv}(A)}| &\leq |(k-e)\text{Conv}(\hat{A})| \leq |\{n \in S_\sigma : n_{d+1} = k-e\}| \leq \\ &\leq |\{n \in m + S_\sigma : n_{d+1} = k\}| \\ &\leq |k \cdot \hat{A}| = \underline{|k \cdot A|} \end{aligned}$$

Since $\underline{\quad} = E_A(k-e)$ and $\underline{\quad} = HF_A(k)$, we
 $\rightarrow$ recall this is always true

have $E_A(k-e) \leq HF_A(k) \leq E_A(k)$

For $k \gg 0$, $HF_A(k) = HP_A(k)$, and the inequality implies that $HP_A(k)$ and $E_A(k)$ have the same leading monomial

Indeed, if $f(x) = \alpha x^k + f'(x)$ $\deg f' < k$
 $g(x) = \beta x^m + g'(x)$ $\deg g' < m$; $\beta > 0$
 $e \in M$ s.t. $g(x-e) \leq f(x) \leq g(x)$ $\forall x \gg 0$

$$\Rightarrow \lim_{x \rightarrow +\infty} \frac{g(x-e)}{g(x)} \leq \frac{f(x)}{g(x)} \leq \frac{g(x)}{g(x)}$$

$$\Rightarrow \lim_{x \rightarrow +\infty} \frac{f(x)}{g(x)} = 1 \Rightarrow \deg f = \deg(g) \text{ and } \alpha = \beta$$

Rmks: i) What if $\text{rk}(\mathbb{Z}A) = d$ but $\mathbb{Z}'A \subsetneq \mathbb{Z}^d$?

Then $\mathbb{Z}^d / \mathbb{Z}'A$ is a finite group of order $:= [\mathbb{Z}^d : \mathbb{Z}'A]$ (called **index** of $\mathbb{Z}'A \subseteq \mathbb{Z}^d$)

$$\text{and } \deg X_A = \frac{d! \cdot \text{Vol}(A)}{[\mathbb{Z}^d : \mathbb{Z}'A]}$$

(see lemma 3.3.4)

ii) What if $\text{rk}(\mathbb{Z}A) < d$? Then there exists a Smith normal form decomposition

$$S = PAQ$$

and for $\bar{A} :=$ "first $\text{rk}(\mathbb{Z}A)$ rows of Q^{-1} ", we

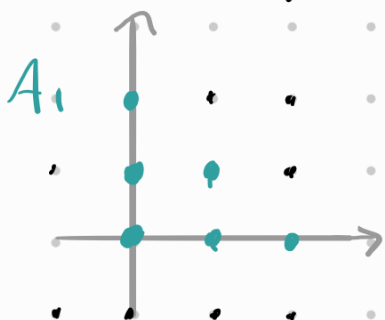
have that $\text{rk}(\mathbb{Z}\bar{A}) = r$, $\mathbb{Z}\bar{A} \subseteq \mathbb{Z}^r$ and

$$\text{im } \psi_A = \text{im } \psi_{\bar{A}}$$

So $\deg X_A = \deg X_{\bar{A}}$ and it is enough to

apply the theorem to $\bar{A}$.

Example : •) $A_1 := \begin{pmatrix} 0 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 2 \end{pmatrix}$, $d=2$

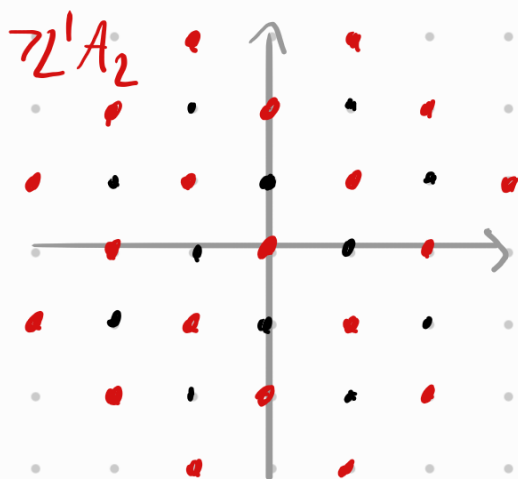
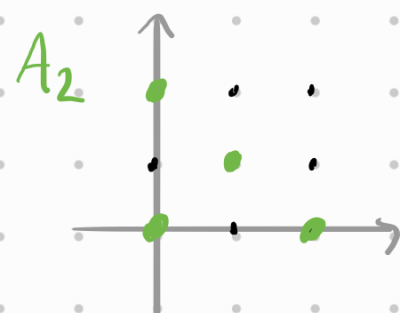


$$\mathbb{Z}^1 A_1 = \mathbb{Z}^2 \quad \checkmark$$

$$\text{Vol}(\text{Conv}(A_1)) = 2$$

$$\text{Kushnirenko} \Rightarrow \deg X_{A_1} = 4$$

•) $A_2 := \begin{pmatrix} 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 2 \end{pmatrix}$



$$\mathbb{Z}^2 / \mathbb{Z}^1 A_2$$



$$\text{Kushnirenko} \Rightarrow \deg X_{A_2} = 2$$

Projective Normality and smoothness

Recall: An affine variety $Y \subseteq \mathbb{A}^n$ is normal if $\mathbb{A}[Y]$ is normal, and:

Def: A variety $X \subseteq \mathbb{P}^n$ is normal if there exists an open covering of affine normal varieties.

Recall:

Thm 2.4.2: If $Y = \text{Spec}(\mathbb{A}[S])$ s.t. $S \subseteq \mathbb{Z}^d$ is an affine semigroup with $\mathbb{Z}S = \mathbb{Z}^d$, then

Y is normal $\Leftrightarrow S$ is saturated

if $K \in \mathbb{N} \setminus \{0\}$, $m \in S$ s.t. $Km \in S$ then $m \in S$.

For a projective toric variety X_A , $A \in \mathbb{Z}^{d \times n}$, we have that $\text{Spec}(\mathbb{A}[\mathbb{N}(A - a_i)])$, i.e. $V(\text{Conv}(A))$ is an affine open cover of Y_A , so

X_A normal $\Leftrightarrow \mathbb{N}(A - a_i)$ is saturated in $\mathbb{Z}(A - a_i)$
 $\forall a_i$ vertex of $\text{Conv}(A)$

Smoothness: Also smoothness is a local property (i.e. it can be checked at any open affine cover)

Hence (Thm 2.4.6 + above discussion) $\Rightarrow$

Assume γ_A is normal. Then

X_A smooth $\Leftrightarrow \text{Cone}(A - a_i)$ is smooth
 $\forall a_i$ vertex of $\text{Conv}(A)$

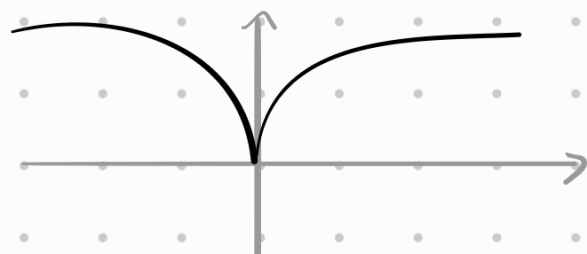
E.g: Consider $A = (0, 2, 3)$ and the corresponding

proj. toric variety $X_A \subseteq \mathbb{P}^2$

$$I(\gamma_A) = (x_1^3 - x_2^2 x_0) \in \mathbb{C}[x_0, x_1, x_2]$$

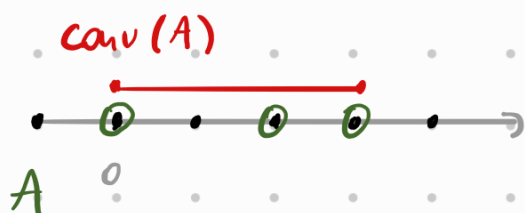
This is not smooth, since $X_A \cap U_0$ $U_0 := \mathbb{P}^2 \setminus V(x_0)$

is the cuspidal cubic $y^3 - x^2 = 0$



some real points of
 $X_A \cap U_0$

We can verify this using the above criterion:



$$A - a_1 = (2, 3)$$

$$\text{Cone}(A - a_1) = \mathbb{N} \setminus \{0\}$$

Minimal generators are $\{2, 3\}$, and they are not
 a subset of a basis of $\mathbb{Z} \Rightarrow X_A$ is not smooth.
 Thm 2.4.6

(Normal) Toric Variety from a polytope

Given a lattice polytope $P \subseteq M_{\mathbb{R}}$, we want to define a normal toric variety X_P associated to P .

The obvious choice $X_{P \cap \mathbb{Z}^d}$ does not always yield a normal variety (e.g. $\text{Conv}(2,3) \subseteq \mathbb{Z} \otimes_{\mathbb{Z}} \mathbb{R}$).

Recall:

$\forall A$ normal $\Leftrightarrow \Pi(A - a_i)$ is saturated in $\mathbb{Z}(A - a_i)$,
 $\forall a_i$ vertex of $\text{Conv}(A)$

Let A be the matrix of the points in $P \cap \mathbb{Z}^d$. For X_A to be normal we need that

Very Ample

$\forall v \in V(P)$, the affine semigroup
 $\Pi \cdot (P \cap M - v)$ is saturated

Hence, P is very ample $\Leftrightarrow X_{P \cap M}$ is normal

Luckily, from a full dimensional polytope we can always obtain a very ample one:

Thm 3.5.6 $P \subseteq M_{\mathbb{R}} \cong \mathbb{R}^d$ s.t. $\dim P = d$. For $K \in \mathbb{N}_{>0}$ let

$K \cdot P := \{p_1 + \dots + p_k : p_i \in P \forall i=1, \dots, k\}$. Then $\forall K \geq d-1$ the K -dilation $K \cdot P$ is very ample.

Def The projective toric variety associated to a full dimensional polytope $P \in M_{\mathbb{R}}$ is isomorphic to $X_{(K \cdot P) \cap M}$, for any $K \in \mathbb{N}$ s.t. $K \cdot P$ is very ample, we denote it χ_P .

Prop: The above definition is well-posed, i.e. $\forall K, \ell$ s.t. $K \cdot P$ and $\ell \cdot P$ are very ample, we have that $X_{(K \cdot P) \cap M} \cong X_{(\ell \cdot P) \cap M}$.

Idea of proof:

i) $\forall K, \ell$ s.t. $K \cdot P$ and $\ell \cdot P$ are very ample we

have $\mathbb{N}((K \cdot P) \cap M - Kv) = \mathbb{N}((\ell \cdot P) \cap M - \ell v)$,

$\forall v \in V(P)$. (Lemma 3.5.11)

$\Rightarrow$ The affine open charts of $X_{(K \cdot P) \cap M}$ and $X_{(\ell \cdot P) \cap M}$ are isomorphic

ii) Find explicit isomorphisms of affine open charts that agree on intersections

For this, let $[L]$ be the matrix of the lattice points in L , $\forall L \in M$

Since by i): $\#((k \cdot P) \cap M - kv) = \#((e \cdot P) \cap M - ev)$,

we have that $\exists A_{k \rightarrow e}^v \in M^{E_P(k) \times E_P(e)}$,

$$\exists A_{e \rightarrow k}^v \in M^{E_P(e) \times E_P(k)}$$

$$\text{s.t. } [(k \cdot P) \cap M - kv] \cdot A_{k \rightarrow e}^v = [(e \cdot P) \cap M - ev]$$

$$\text{and } [(e \cdot P) \cap M - ev] \cdot A_{e \rightarrow k}^v = [(k \cdot P) \cap M - kv]$$

For each $v \in V(P)$, the vertex $k \cdot v$ of $k \cdot P$

corresponds to a coordinate on $\mathbb{P}^{E_P(k)-1}$. Let

$U_{k,v}$ be the principal open subset where this

coordinate does not vanish. Similarly define

$$U_{e,v} \subseteq \mathbb{P}^{E_P(e)-1}.$$

Let $\varphi_{k \rightarrow e}^v : U_{k,v} \rightarrow U_{e,v}$ be defined

by

$$(x_m)_{m \in k \cdot P \cap M} \mapsto \varphi_{k \rightarrow e}^v \left[\left(\frac{x_m}{x_{kv}} \right)_{m \in k \cdot P \cap M} \right]$$

And similarly $\varphi_{e \rightarrow k}^v : U_{e,v} \rightarrow U_{k,v}$.

These are

- i) morphisms of affine varieties (i.e. regular)
- ii) they agree on intersections, i.e.

if $v, w \in V(P)$ and $x \in X_{(k,P) \cap M} \cap U_{k,v} \cap U_{k,w}$

then $\varphi_{k \rightarrow e}^v(x) = \varphi_{k \rightarrow e}^w(x)$ (lemma 3.5.13)

$\Rightarrow$ can define $\varphi_{k \rightarrow e}(x) : \mathbb{P}^{E_P(k)-1} \rightarrow \mathbb{P}^{E_P(e)-1}$

by $\varphi_{k \rightarrow e}|_{U_{k,v}} := \varphi_{k \rightarrow e}^v, \forall v \in V(P)$

This turn out to be isomorphisms between

$X_{[(k,P) \cap M]}$ and $X_{[(e,P) \cap M]}$.

